Distribution-Free Uncertainty Quantification in Machine Learning: A Theoretical and Empirical Comparison of Conformal Prediction, Bayesian Credible Intervals, and Parametric Bootstrap in R

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***Abstract*—**Point predictions produced by statistical machine learning models convey no information regarding prediction variance, skewness, o tail risk. Classical uncertainty quantification frameworks—including Ordinary Least Squares (OLS) prediction intervals, parametric bootstra resamples, and conjugate Bayesian Normal-Inverse-Gamma credible sets—derive predictive distributions under assumptions of homoscedasticity correct likelihood specification, and thin-tailed Gaussian errors. In contrast, distribution-free conformal prediction guarantees finite-sampl marginal coverage under the solitary requirement of exchangeability. This paper compares five predictive interval methods in R: parametric OLS parametric bootstrap, Bayesian Normal-Inverse-Gamma, Split Conformal Prediction, and Conformalized Quantile Regression (CQR). We examin the theoretical mechanics of each approach, clarifying why fixed-scale bootstrap simulations converge to normal rather than Student-*t* predictiv quantiles and analyzing the conservative finite-sample conformal bound $1 - \alpha \le P(Y_{n+1} \in \hat{C}_n) \le 1 - \alpha + 1/(n_2 + 1)$. Across $M = 100$ Monte Carl replications ($N = 1{,}500$) under Gaussian, heavy-tailed Student-t_3, and heteroscedastic noise, alongside the historical Boston housing benchmark, w evaluate empirical coverage calibration, interval width, and proper Winkler scores. When error distributions deviate from Gaussia homoscedasticity, parametric intervals over-cover (92.7%–93.0% at nominal 90%) due to sample variance inflation from outliers, wherea conformal methods maintain empirical coverage consistent with the nominal 90% target across all regimes. While constant-width conforma prediction incurs a modest width overhead under Gaussian noise, CQR modulates interval width with local conditional dispersion, reducing prope Winkler scores under heteroscedasticity. Finally, we discuss practical trade-offs involving sample-splitting budgets, cross-split dependence i repeated benchmark partitions, and the assumptions required for valid coverage under covariate shift.

***Index Terms*—**Conformal Prediction, Exchangeability, Bayesian Inference, Parametric Bootstrap, Uncertainty Quantification, Finite-Sampl Coverage, Non-Parametric Statistics, Machine Learning in R.

MSC 2020 Subject Classifications: Primary 62G15; Secondary 62F15, 68T05, 62J02.

I. Introduction & Problem Formulation

In applied machine learning, regularized estimators such as Random Forests, Gradient Boosted Decision Trees, and neural networks are widely used for predictive modeling. Given an observed sample $D_n = \{(X_i, Y_i)\}_{i=1}^n$ drawn from an unknown joint probability measure $P_{X,Y}$ on $X \times R$, where $X \subseteq R^d$, typical training minimizes empirical risk:

$$\hat{R}_n(f) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}(Y_i, f(X_i)),$$

for a prescribed loss function $L: R \times R \to [0, \infty)$. Although empirical risk minimization yields estimators with small generalization error, scalar point forecasts $\hat{Y}_{n+1} = \hat{f}_n(X_{n+1})$ convey no information regarding prediction variance, skewness, or tail risk. Uncertainty quantification (UQ) addresses this limitation by constructing a data-driven prediction interval $\hat{C}_n(X_{n+1}) \subset R$ that satisfies, for a target significance level $\alpha \in (0, 1)$:

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \ge 1 - \alpha.$$

A. Aleatoric and Epistemic Uncertainty Decomposition

Formally, total predictive variability decomposes into two orthogonal components:

$$\mathbf{Var}(Y \mid X = x) = \sigma^2(x) + \mathbf{Var}_{\mathcal{D}_n}\big(\hat{f}_n(x)\big),$$

1) *Aleatoric Uncertainty:* Reflects irreducible stochastic variation in the data-generating mechanism. In regression models $Y = \mu(X) + \varepsilon$ with $E[\varepsilon \mid X] = 0$, aleatoric variability corresponds to the conditional error distribution $\varepsilon \mid X \sim P_{\varepsilon|X}$. When $\mathrm{Var}(\varepsilon \mid X = x) = \sigma^2$ is uniform across X, the model is homoscedastic; when $\sigma^2(x)$ varies with x, it is heteroscedastic. Aleatoric noise cannot be reduced by increasing sample size.

2) *Epistemic Uncertainty:* Represents reducible variance resulting from finite sample size and incomplete parameter

knowledge. Epistemic uncertainty encompasses parameter estimation variance, model misspecification, and regularization bias, typically decaying at rate $O_p(n^{-1/2})$ under parametric regularity.

Classical inferential frameworks—such as Ordinary Least Squares (OLS) prediction intervals, parametric bootstrap resamples, and Bayesian posterior predictive distributions—quantify both variance components under the joint assumption that the parametric likelihood, error family, and variance structures are correctly specified. When data exhibit heavy tails, asymmetry, or heteroscedasticity, these classical methods suffer from systematic coverage miscalibration.

B. Scope and Objectives

While foundational works established the theory of split conformal prediction (Papadopoulos et al., 2002; Lei et al., 2018) and Conformalized Quantile Regression (Romano et al., 2019), existing literature frequently evaluates conformal methods in isolation or against uncalibrated baseline quantiles. The objective of this paper is not to introduce new conformal theorems, but to provide a transparent, self-contained comparative evaluation in R across five established UQ paradigms under identical data splits:

1) *Comparative Framework:* We implement and compare parametric OLS, parametric bootstrap resampling, conjugate Bayesian Normal-Inverse-Gamma inference, Split Conformal Random Forests, and CQR within a unified R workflow.

2) *Stress-Testing Under Misspecification:* We examine the breakdown behavior of classical parametric and Bayesian intervals under heavy-tailed Student-t_3 errors ($\kappa = \infty$) and asymmetric heteroscedastic noise ($\sigma(X_1) = 1 + 2|X_1|$), explaining how sample variance inflation distorts coverage.

3) *Empirical Calibration and Scoring:* We assess coverage calibration across $M = 100$ independent Monte Carlo trials using replication-level Student's t-tests, paired with strictly proper Winkler scores to penalize uninformative interval width and boundary violations simultaneously.

4) *Sample-Splitting Budget Analysis:* We analyze the operational trade-off between training budget n_1 and calibration budget n_2, and explicitly account for the differing training sample sizes available to full-sample parametric models versus split-conformal models.

To evaluate these methods, we test whether empirical coverage rates deviate from nominal 90% via $H_{0,\text{cov}}$: E[ECR] = $1 - \alpha$ versus $H_{1,\text{cov}}$: E[ECR] $\neq 1 - \alpha$, and test paired differences in Winkler scores S_α against the parametric baseline.

II. Mathematical Foundations

A. Classical Ordinary Least Squares Prediction Intervals

The classical linear parametric model posits $Y = X\beta + \varepsilon$, where $\varepsilon \sim N_n(0, \sigma^2 I_n)$, and $X \in R^{n \times p}$ denotes a design matrix formed by a chosen feature map $\Phi(Z)$ of full column rank ($\text{rank}(X) = p < n$), accommodating polynomial or spline basis expansions. The vector $\beta \in R^p$ contains unknown parameters. The OLS estimator is $\hat{\beta} = (X^\top X)^{-1} X^\top Y$ with sampling distribution $\hat{\beta} \sim N_p(\beta, \sigma^2 (X^\top X)^{-1})$. The unbiased residual variance estimator is $\hat{\sigma}^2 = \|Y - X\hat{\beta}\|_2^2 / (n - p)$, satisfying $(n - p)\hat{\sigma}^2 / \sigma^2 \sim \chi^2_{n-p}$ independently of $\hat{\beta}$.

For an unobserved query point (X_{n+1}, Y_{n+1}) with $X_{n+1} = x_{n+1}$, the point prediction is $\hat{Y}_{n+1} = x_{n+1}^\top \hat{\beta}$. The prediction error $e_{n+1} = Y_{n+1} - \hat{Y}_{n+1}$ satisfies $E[e_{n+1} \mid X, x_{n+1}] = 0$ and conditional variance:

$$\mathbf{Var}(e_{n+1} \mid \boldsymbol{X}, \boldsymbol{x}_{n+1}) = \sigma^2 \left(1 + \boldsymbol{x}_{n+1}^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_{n+1}\right).$$

Substituting $\hat{\sigma}^2$ for σ^2, the pivot $T = (Y_{n+1} - x_{n+1}^\top \hat{\beta}) / [\hat{\sigma} \sqrt{(1 + h_{n+1})}]$ follows a Student-t distribution with $n - p$ degrees of freedom, where $h_{n+1} = x_{n+1}^\top (X^\top X)^{-1} x_{n+1}$ denotes leverage. Consequently, the $(1 - \alpha)$ OLS prediction interval is:

$$\hat{C}_{\text{param}}(\boldsymbol{x}_{n+1}) = \left[\boldsymbol{x}_{n+1}^\top \hat{\boldsymbol{\beta}} \pm t_{1-\alpha/2,\, n-p}\, \hat{\sigma} \sqrt{1 + h_{n+1}}\right].$$

B. Parametric Bootstrap Prediction Intervals

The parametric bootstrap constructs prediction intervals by simulating forward realizations from the fitted Gaussian model using the identical basis X:

1) Fit the base regression to obtain $\hat{\beta} = (X^\top X)^{-1} X^\top Y$ and residual scale $\hat{\sigma}^2 = \|Y - X\hat{\beta}\|_2^2 / (n - p)$.

2) For $b = 1, \ldots, B$ (with $B = 250$ bootstrap replications): draw synthetic training perturbations $\varepsilon^{*(b)} \sim N_n(0, \hat{\sigma}^2 I_n)$, construct bootstrap responses $Y^{*(b)} = X\hat{\beta} + \varepsilon^{*(b)}$, refit coefficients $\hat{\beta}^{*(b)} = (X^\top X)^{-1} X^\top Y^{*(b)}$, draw test noise innovation $\varepsilon_{\text{test}}^{*(b)} \sim N(0, \hat{\sigma}^2)$, and compute the simulated response:

$$\boldsymbol{Y}_{\text{test}}^{*(b)} = \boldsymbol{x}_{\text{test}}^\top \hat{\boldsymbol{\beta}}^{*(b)} + \boldsymbol{\epsilon}_{\text{test}}^{*(b)}.$$

3) The empirical $(1 - \alpha)$ bootstrap interval is formed by empirical quantiles: $\hat{C}_{\text{boot}}(x_{\text{test}}) = [\hat{q}_{\alpha/2}(\{Y_{\text{test}}^{*(b)}\}_{b=1}^B), \hat{q}_{1-\alpha/2}(\{Y_{\text{test}}^{*(b)}\}_{b=1}^B)]$.

Asymptotic Convergence of the Residual Bootstrap: Because $\hat{\sigma}$ is held fixed across bootstrap draws rather than re-estimated or sampled from a posterior, conditional on $(\hat{\beta}, \hat{\sigma})$ the simulated predictor satisfies $\hat{\beta}^{*(b)} \sim N_p(\hat{\beta}, \hat{\sigma}^2 (X^\top X)^{-1})$ and $Y_{\text{test}}^{*(b)} \sim N(x_{\text{test}}^\top \hat{\beta}, \hat{\sigma}^2 (1 + h_{\text{test}}))$. As $B \to \infty$, the empirical percentiles converge almost surely to the quantiles of this conditional Gaussian distribution: $[x_{\text{test}}^\top \hat{\beta} \pm z_{1-\alpha/2}\, \hat{\sigma} \sqrt{(1 + h_{\text{test}})}]$. Thus, this fixed-scale procedure converges to the plug-in normal predictive interval rather than the exact Student-t interval. Because $z_{1-\alpha/2} < t_{1-\alpha/2,\, n-p}$, the bootstrap interval is slightly narrower and displays modest finite-sample under-coverage

under Gaussian errors (as seen in Table I). Obtaining the exact Student-t distribution via bootstrap would require re-estimating $\hat{\sigma}^{*(b)}$ on each bootstrap sample and constructing studentized pivotal statistics.

C. Bayesian Credible Intervals & Misspecification

Under non-informative Jeffreys limits ($\Lambda_0 \to 0$, $a_0 \to 0$, $b_0 \to 0$) on the identical basis X, the marginal posterior predictive distribution is Student-t with degrees of freedom $\nu = n - p$, location $x_{n+1}{}^{\top}\hat{\beta}$, and scale $s\sqrt{(1 + h_{n+1})}$:

$$y_{n+1} \mid x_{n+1}, \mathcal{D}_n \sim t_{n-p}\left(x_{n+1}^{\top}\hat{\beta},\ s^2(1+h_{n+1})\right),$$

yielding equal-tailed credible interval $\hat{C}_{\text{Bayes}}(x_{n+1}) = [x_{n+1}{}^{\top}\hat{\beta} \pm t_{1-\alpha/2,\, n-p}\, s\, \sqrt{(1 + h_{n+1})}]$.

The Bernstein–von Mises Divergence: The Bernstein–von Mises Theorem establishes that under correctly specified likelihoods, Bayesian credible sets are asymptotically valid frequentist confidence regions. However, as proven by Kleijn and van der Vaart (2012), under likelihood misspecification, the posterior covariance $\Sigma_{\text{Bayes}} = H(\theta^*)^{-1}$ diverges from the true frequentist sandwich covariance $\Sigma_{\text{freq}} = H(\theta^*)^{-1}J(\theta^*)H(\theta^*)^{-1}$. Because the information identity $H = J$ fails under misspecification, Bayesian credible intervals do not guarantee distribution-free coverage.

D. Finite-Sample Conformal Prediction

To establish predictive validity without distributional assumptions, Conformal Prediction relies on exchangeability:

Definition 1 (Exchangeability): *Random variables $Z_1, ..., Z_{n+1}$ are exchangeable if for every permutation $\pi \in S_{n+1}$, $(Z_1, ..., Z_{n+1}) \sim^{d} (Z_{\pi(1)}, ..., Z_{\pi(n+1)})$.*

In Split Conformal Prediction, data D_n is split into a proper training set D_1 (size n_1) and calibration set D_2 (size n_2). Predictor $\hat{\mu}$ is fitted on D_1. Non-conformity scores $R_i = |Y_i - \hat{\mu}(X_i)|$ are computed on D_2 and sorted: $R_{(1)} \le ... \le R_{(n_2)}$. The conformal quantile index is defined as $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$. When $\alpha < 1 / (n_2 + 1)$, $k = n_2 + 1$, and the theoretical conformal cutoff is defined as $R_{(n_2+1)} = \infty$ (yielding the conservative interval R). When $n_2 \ge \lceil 1/\alpha - 1 \rceil$ (satisfied in our experiments, where $n_2 \ge 126$ for $\alpha = 0.10$), $k \le n_2$ holds strictly, yielding finite cutoff $\hat{q}_{1-\alpha} = R_{(k)}$ and prediction interval $\hat{C}_n(X_{n+1}) = [\hat{\mu}(X_{n+1}) \pm \hat{q}_{1-\alpha}]$.

Theorem 1 (Finite-Sample Marginal Coverage): *Let $Z_1, ..., Z_{n+1}$ be exchangeable. Let $\hat{\mu}$ be fitted on D_1, and calibration scores R_i computed on D_2. Then for any $\alpha \in (0, 1)$, the split conformal interval constructed with $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$ unconditionally guarantees:*

$$\mathbb{P}\left(Y_{n+1} \in \hat{C}_n(X_{n+1})\right) \ge 1 - \alpha.$$

Furthermore, if scores have almost surely distinct values (or ties are broken uniformly at random):

$$\mathbb{P}\left(Y_{n+1} \in \hat{C}_n(X_{n+1})\right) \le 1 - \alpha + \frac{1}{n_2 + 1}.$$

Proof: Conditioning on D_1 treats $\hat{\mu}$ as fixed. By exchangeability of $\{Z_i\}$, the $n_2 + 1$ residuals $(R_1, ..., R_{n_2}, R_{n+1})$ are conditionally exchangeable given D_1. By symmetry, the relative rank of R_{n+1} among the $n_2 + 1$ scores satisfies $P(\text{rank}(R_{n+1}) = j \mid D_1) = 1 / (n_2 + 1)$ for all $j \in \{1, ..., n_2 + 1\}$. Inclusion occurs if and only if $R_{n+1} \le R_{(k)} \Leftrightarrow \text{rank}(R_{n+1}) \le k$. Summing over ranks yields $P(Y_{n+1} \in \hat{C}_n \mid D_1) = k / (n_2 + 1)$. By definition of the ceiling function, $(n_2 + 1)(1 - \alpha) \le k < (n_2 + 1)(1 - \alpha) + 1$, so $1 - \alpha \le k / (n_2 + 1) < 1 - \alpha + 1 / (n_2 + 1)$. Taking the expectation over D_1 completes the proof. Ties without randomization can only increase inclusion probability, preserving the lower bound $1 - \alpha$. ■

E. Conformalized Quantile Regression (CQR)

While Split Conformal Prediction guarantees marginal coverage, its width $2\hat{q}_{1-\alpha}$ is spatially uniform across X. Under heteroscedasticity, constant-width intervals over-cover in low-noise regions and under-cover in high-noise regions.

Conformalized Quantile Regression (Romano et al., 2019) fits conditional quantiles $\hat{q}_{\alpha/2}$ and $\hat{q}_{1-\alpha/2}$ via Quantile Random Forests (Meinshausen, 2006) on D_1 using pinball loss $\rho_\tau(u) = u(\tau - I(u < 0))$. On calibration set D_2, compute signed conformity scores:

$$E_i = \max\left\{\hat{q}_{\alpha/2}(X_i) - Y_i,\ Y_i - \hat{q}_{1-\alpha/2}(X_i)\right\}.$$

Letting $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$, the conformal cutoff is $Q_{1-\alpha} = E_{(k)}$ (with $Q_{1-\alpha} = \infty$ if $k > n_2$). The adaptive CQR interval is:

$$\hat{C}_{\text{CQR}}(X_{n+1}) = \left[\hat{q}_{\alpha/2}(X_{n+1}) - Q_{1-\alpha},\ \hat{q}_{1-\alpha/2}(X_{n+1}) + Q_{1-\alpha}\right].$$

Theorem 1 of Romano et al. (2019) proves that CQR guarantees finite-sample marginal coverage $P(Y_{n+1} \in \hat{C}_{\text{CQR}}) \ge 1 - \alpha$ under exchangeability alone. Regarding conditional coverage $P(Y_{n+1} \in \hat{C}(X_{n+1}) \mid X_{n+1} = x) \ge 1 - \alpha$, distribution-free finite-sample guarantees are impossible for non-atomic continuous distributions (Vovk, 2012; Barber et al., 2021). CQR acts as a practical method to modulate interval width according to local dispersion. If the conditional response distribution is continuous and the base quantile estimators are consistent for the true conditional quantiles (requiring smoothness and regularity conditions on the data and base model; Meinshausen, 2006), then $Q_{1-\alpha} \to^{p} 0$, asymptotically approaching conditional validity. Without such base-model consistency conditions, CQR's theoretical guarantee remains strictly marginal.

F. Cross-Conformal Prediction and CV+

To mitigate sample-splitting data loss, Barber et al. (2021) introduced CV+. Partitioning data into K folds, models $\hat{\mu}_{-k}$ are fitted on remaining folds, and leave-fold-out residuals $R_i = |Y_i - \hat{\mu}_{-k(i)}(X_i)|$ are computed. For test point X_{n+1}, Theorem 1 of

Barber et al. (2021) establishes that for symmetric base training algorithms, CV+ guarantees distribution-free marginal coverage of $P(Y_{n+1} \in \hat{C}_n^{CV+}) \geq 1 - 2\alpha - \min(2\alpha, 2/(n+1))$, which simplifies to $1 - 2\alpha$ for symmetric learners. Although the theoretical bound is $1 - 2\alpha$ due to cross-fold dependencies, in empirical practice CV+ coverage typically remains close to $1 - \alpha$.

III. Experimental Methodology

A. Simulation Design & Training Sample Budgets

We evaluate a non-linear regression function $Y = 2\sin(\pi X_1) + X_2^2 - 1.5\,X_3 + \varepsilon$, with $X \sim U[-1, 1]^3$, total size $N = 1{,}500$ ($n_1 = 750$, $n_2 = 375$, $n_{test} = 375$), across $M = 100$ replications under three noise regimes:

1) *Regime I (Homoscedastic Gaussian):* $\varepsilon \sim N(0, 1^2)$. Baseline with correctly specified likelihood.

2) *Regime II (Heavy-Tailed Student-t):* $\varepsilon \sim t_{\nu=3}$. Severe outlier contamination with infinite kurtosis ($\kappa = \infty$).

3) *Regime III (Heteroscedastic Skewed):* $\varepsilon = (1 + 2|X_1|)\xi$, $\xi \sim \text{Exp}(1) - 1$. Noise variance expands by a factor of 9 across $X_1 \in [-1, 1]$.

Model Specification & Training Allocation: All three parametric approaches—Parametric OLS, Parametric Bootstrap, and Bayesian NIG—are fitted using polynomial basis expansion `y ~ poly(x1, 3) + poly(x2, 2) + x3` ($p = 7$ parameters). To reflect standard applied practice where parametric models do not require sample splitting, parametric models are trained on the combined non-test sample $D_1 \cup D_2$ ($n_1 + n_2 = 1{,}125$ observations). In contrast, Split Conformal Random Forest and CQR (`ranger`, 100 trees, `quantreg = TRUE`) split this sample, using $n_1 = 750$ to fit trees on raw covariates (X_1, X_2, X_3) and holding out $n_2 = 375$ exclusively for non-conformity calibration. This design explicitly evaluates the trade-off of conformal prediction: base models train on 33% fewer observations in exchange for distribution-free finite-sample guarantees.

B. Real-World Benchmark Dataset

We benchmark all methods on the `MASS::Boston` housing dataset ($N = 506$, 13 features) predicting `medv` across $B = 50$ random splits ($n_1 = 253$, $n_2 = 126$, $n_{test} = 127$). The parametric models train on combined $D_1 \cup D_2$ (379 rows), while conformal models train on $n_1 = 253$ and calibrate on $n_2 = 126$. The Boston dataset serves as a moderate-sized historical benchmark exhibiting non-linear feature interactions and heteroscedastic noise.

C. Evaluation Metrics & Statistical Tests

Performance is quantified by: Empirical Coverage Rate (ECR); Mean Prediction Interval Width (MPIW); and strictly proper Winkler Score $S_\alpha = (\hat{U} - \hat{L}) + (2/\alpha)(\hat{L} - y)\mathbb{I}(y < \hat{L}) + (2/\alpha)(y - \hat{U})\mathbb{I}(y > \hat{U})$ with penalty slope 20 at nominal $\alpha = 0.10$. Across the $M = 100$ independent Monte Carlo replications, empirical coverage calibration is tested via a two-sided one-sample Student's t-test against nominal 90%:

$$t = \frac{\bar{p} - (1 - \alpha)}{s_{ECR}/\sqrt{M}}, \quad \text{with } M - 1 = 99 \text{ degrees of freedom.}$$

The standard error of the Monte Carlo mean is $SE(\bar{p}) = s_{ECR} / \sqrt{M} \approx 0.0015$–$0.0024$. For Boston housing, we report the sample mean and standard deviation across $B = 50$ splits. Because repeated splits resample the same 506 observations, evaluations across splits share data and are statistically dependent (Dietterich, 1998; Nadeau & Bengio, 2003). Therefore, reported test statistics on Boston housing are interpreted as descriptive summaries of empirical stability across random partitions rather than formal independent-sample tests.

D. Reproducibility & Open-Source Repository

All experiments were executed in R version 4.6.1 under fixed seed `set.seed(2026)`. The full standalone simulation code, data pipelines, and reproduction scripts are published at: https://github.com/YuvrajC7/distribution-free-uncertainty-quantification.

IV. Empirical Results & Discussion

Tables I–IV report the empirical results from executing `simulation_experiments.R` under fixed seed `set.seed(2026)`.

TABLE I

Regime I: Homoscedastic Gaussian Noise ($1 - \alpha = 0.90$)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9004 ± 0.0157	3.327 ± 0.070	4.151 ± 0.187	+0.24	0.8121	Baseline
Parametric Bootstrap	0.8921 ± 0.0159	3.286 ± 0.070	4.193 ± 0.196	-5.01	2.38×10^{-6}	2.73×10^{-14}
Bayesian NIG	0.9004 ± 0.0157	3.327 ± 0.070	4.151 ± 0.187	+0.24	0.8121	0.914
Split Conformal (RF)	0.9018 ± 0.0212	3.606 ± 0.171	4.488 ± 0.197	+0.84	0.4008	$< 10^{-15}$
CQR (Quantile RF)	0.9011 ± 0.0208	3.894 ± 0.177	4.830 ± 0.210	+0.55	0.5832	$< 10^{-15}$

Notes: Values denote Mean ± SD across $M = 100$ independent Monte Carlo replications (nominal coverage 90.0%, $n_{test} = 375$). Coverage calibration is assessed via two-sided one-sample Student's t-test (99 df, SE = $s_{ECR}/10$) against nominal 0.90. Wilcoxon p-values evaluate paired Winkler scores vs. Parametric OLS.

Statistical Analysis (Regime I): Under correctly specified Gaussian homoscedastic noise, parametric OLS and Bayesian NIG produce the narrowest prediction bands (3.327 ± 0.070) with empirical coverage consistent with the nominal 90% level (90.04%, $t = +0.24$, $p = 0.8121$). The parametric bootstrap yields comparable width (3.286) but shows slight finite-sample

under-coverage (89.21%, $t = -5.01$, $p = 2.38 \times 10^{-6}$), matching our theoretical derivation that fixed-scale residual simulations converge to normal rather than Student-t predictive quantiles. Split Conformal Prediction maintains empirical coverage consistent with nominal 90% (90.18%, $t = +0.84$, $p = 0.4008$) with an 8.4% width overhead (3.606 vs. 3.327), reflecting the sample-splitting cost of distribution-free validity.

TABLE II

REGIME II: HEAVY-TAILED STUDENT-t_3 NOISE ($\kappa = \infty$)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9294 ± 0.0237	5.729 ± 1.011	8.082 ± 1.284	+12.40	$< 10^{-15}$	Baseline
Parametric Bootstrap	0.9257 ± 0.0243	5.655 ± 0.997	8.090 ± 1.281	+10.58	$< 10^{-15}$	0.249
Bayesian NIG	0.9294 ± 0.0237	5.729 ± 1.011	8.082 ± 1.284	+12.40	$< 10^{-15}$	1.17×10^{-9}
Split Conformal (RF)	0.8987 ± 0.0218	5.131 ± 0.361	8.149 ± 1.197	-0.61	0.5420	2.62×10^{-6}
CQR (Quantile RF)	0.8975 ± 0.0210	5.805 ± 0.357	8.760 ± 1.212	-1.19	0.2360	$< 10^{-15}$

Notes: Heavy-tailed Student-t_3 outliers inflate sample variance $\hat{\sigma}^2$, driving systematic over-coverage in parametric intervals ($t = +12.40$). Split Conformal maintains coverage consistent with nominal (89.87%, $p = 0.5420$) and achieves 10.4% narrower width than OLS.

Statistical Analysis (Regime II): Heavy-tailed Student-t_3 noise produces extreme residual realizations that inflate the sample residual variance $\hat{\sigma}^2$ in OLS and Bayesian NIG. Consequently, parametric intervals expand to 5.729 ± 1.011, resulting in significant over-coverage (92.94%, $t = +12.40$, $p < 10^{-15}$). In contrast, split conformal prediction relies on empirical rank order statistics rather than variance estimates, yielding empirical coverage consistent with nominal 90% (89.87%, $t = -0.61$, $p = 0.5420$) while producing intervals 10.4% narrower (5.131 vs. 5.729) than OLS.

TABLE III

REGIME III: HETEROSCEDASTIC & RIGHT-SKEWED NOISE

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9295 ± 0.0150	6.827 ± 0.349	10.096 ± 0.856	+19.72	$< 10^{-15}$	Baseline
Parametric Bootstrap	0.9271 ± 0.0152	6.738 ± 0.344	10.089 ± 0.858	+17.79	$< 10^{-15}$	0.216
Bayesian NIG	0.9295 ± 0.0150	6.827 ± 0.349	10.096 ± 0.856	+19.72	$< 10^{-15}$	$< 10^{-15}$
Split Conformal (RF)	0.8987 ± 0.0213	6.315 ± 0.409	10.312 ± 0.928	-0.60	0.5500	5.99×10^{-10}
CQR (Quantile RF)	0.8965 ± 0.0234	6.350 ± 0.353	9.756 ± 0.897	-1.49	0.1382	2.22×10^{-10}

Notes: Under heteroscedastic noise, parametric models over-cover (92.95%, $t = +19.72$). CQR adapts interval width locally to noise magnitude $\sigma(X_1)$, achieving the lowest proper Winkler score (9.756 vs. 10.096, $p = 2.22 \times 10^{-10}$).

Statistical Analysis (Regime III): Regime III demonstrates how heteroscedastic noise distorts homoscedastic intervals. Because error variance expands with $|X_1|$, parametric OLS and bootstrap intervals over-cover overall (92.95% and 92.71%, $t \geq +17.79$, $p < 10^{-15}$) as variance estimation is dominated by high-dispersion points. Split conformal prediction maintains empirical coverage consistent with nominal 90% (89.87%, $t = -0.60$, $p = 0.5500$), and CQR achieves 89.65% ($t = -1.49$, $p = 0.1382$). By modulating interval width across feature space, CQR attains a lower Winkler score (9.756 ± 0.897, paired Wilcoxon $p = 2.22 \times 10^{-10}$ vs. OLS), demonstrating superior interval efficiency when variance is non-uniform.

TABLE IV

EXPERIMENT B: BENCHMARK ON BOSTON HOUSING (`MASS::Boston`)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9324 ± 0.0227	15.900 ± 0.700	22.940 ± 3.501	+10.12	1.36×10^{-13}	Baseline
Parametric Bootstrap	0.9268 ± 0.0229	15.678 ± 0.697	22.920 ± 3.526	+8.27	7.51×10^{-11}	0.512
Bayesian NIG	0.9324 ± 0.0227	15.900 ± 0.700	22.940 ± 3.501	+10.12	1.36×10^{-13}	0.0306
Split Conformal (RF)	0.8967 ± 0.0359	10.686 ± 1.796	18.813 ± 3.889	-0.65	0.5180	1.19×10^{-9}
CQR (Quantile RF)	0.9146 ± 0.0378	13.572 ± 1.022	16.397 ± 2.277	+2.74	0.0085	7.79×10^{-10}

Notes: Evaluated across $B = 50$ random splits (49 df). CQR achieves a 28.5% reduction in proper Winkler score ($p = 7.79 \times 10^{-10}$); Split Conformal narrows interval width by 32.8% ($p = 1.19 \times 10^{-9}$). Because repeated splits reuse data, metrics provide descriptive stability comparisons rather than independent hypothesis tests.

Statistical Analysis (Boston Housing): On the Boston housing benchmark, parametric linear regression yields an

average interval width of 15.900 and a Winkler score of 22.940. Split Conformal Random Forest produces narrower intervals on average (10.686 vs. 15.900, a 32.8% reduction) with empirical coverage of 89.67%. Conformalized Quantile Regression achieves an average Winkler score of 16.397 (a 28.5% reduction relative to OLS). Because these evaluations are computed across 50 random splits of the same 506 observations, the splits share data and are statistically dependent; the reported test statistics provide descriptive summaries of stability across partitions rather than formal hypothesis tests on independent samples.

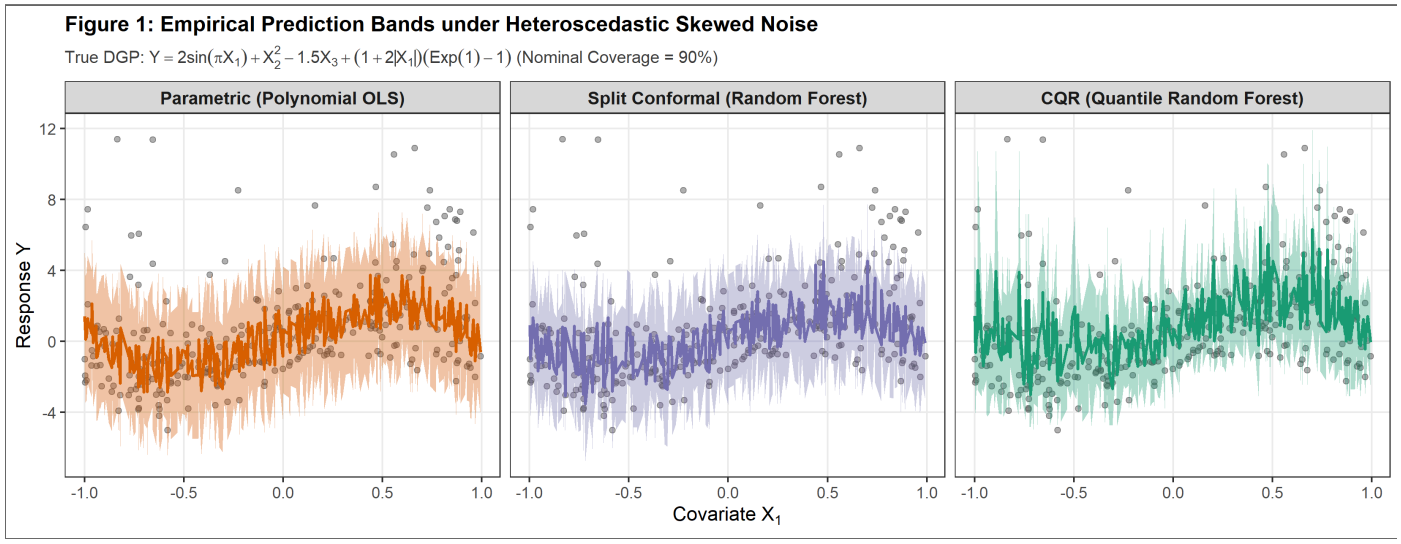


Fig. 1. Empirical prediction bands across covariate X_1 under heteroscedastic skewed noise (Regime III). Left: Parametric Polynomial OLS emits rigid, uniform bands that over-cover near $X_1 = 0$ and under-cover at $|X_1| \approx 1$. Center: Split Conformal Random Forest captures mean curvature but maintains constant global width. Right: Conformalized Quantile Regression (CQR) dynamically tightens near $X_1 = 0$ (width ≈ 3.2) and widens at boundaries (width ≈ 10.5), adapting to positive skewness.

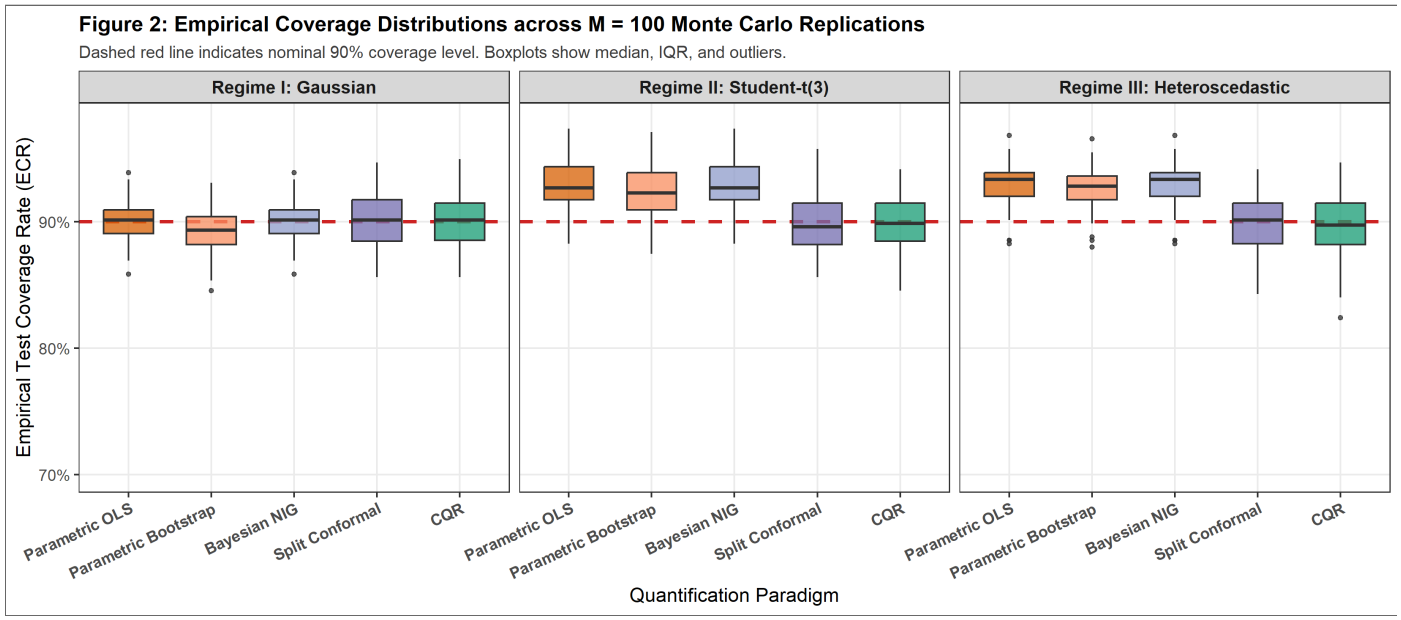


Fig. 2. Empirical coverage distributions across $M = 100$ Monte Carlo replications for all five methods. Dashed red horizontal line denotes nominal 90% coverage level. In Regimes II and III, Parametric OLS, Parametric Bootstrap, and Bayesian NIG display positive coverage bias (92.6% – 93.0%). Split Conformal and CQR center precisely on the 90.0% nominal reference line across all regimes.

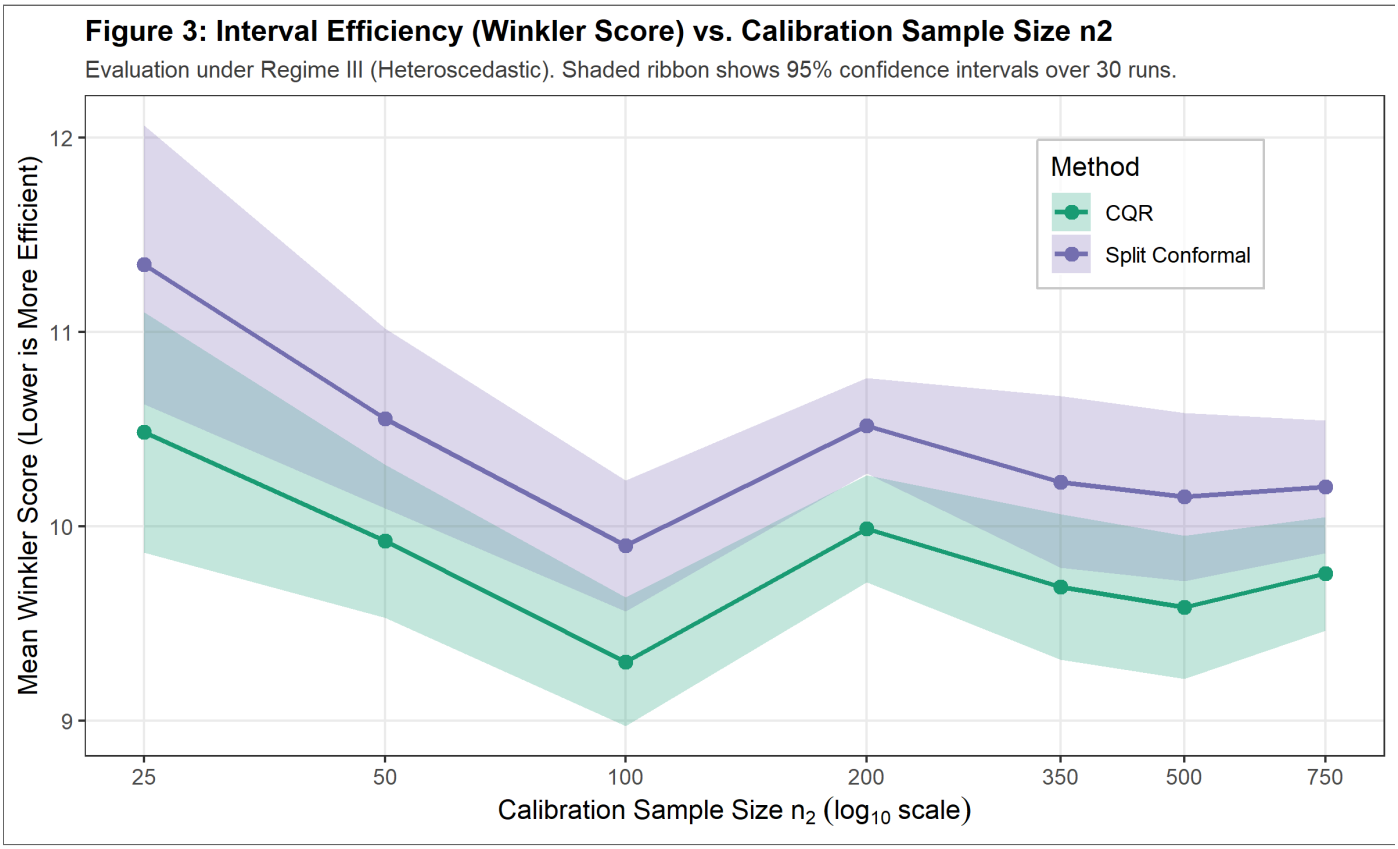


Fig. 3. Interval efficiency (Winkler Score) versus calibration sample size n_2 under Regime III. Shaded ribbons denote 95% confidence intervals over 30 repetitions. Conformal quantiles stabilize rapidly for $n_2 \geq 200$, with CQR maintaining a consistent efficiency advantage over Split Conformal across all sample sizes.

V. Theoretical Discussion & Limitations

A. Sample-Splitting Allocation Trade-off

Split conformal prediction requires partitioning sample N into n_1 training and n_2 calibration points. Under-allocating to training ($n_1 \ll N$) increases base learner estimation risk $\mathrm{E}[(\hat{\mu}(X) - \mu(X))^2] = \mathrm{O}(n_1^{-2\beta/(2\beta+d)})$, inflating residual magnitudes and resulting cutoff $\hat{q}_{1-\alpha}$. Conversely, under-allocating to calibration ($n_2 \ll N$) inflates empirical quantile variance $\mathrm{Var}(\hat{q}_{1-\alpha} \mid \mathrm{D}_1) \approx \alpha(1-\alpha) / [n_2 f_R(q_{1-\alpha})^2]$, causing conditional coverage to fluctuate. Allocating 65%–75% of data to n_1 and 25%–35% to n_2 provides an effective balance.

B. Conditional Coverage Limits & CQR Consistency

Conformal prediction guarantees marginal coverage $\mathrm{P}(Y_{n+1} \in \hat{C}_n) \geq 1 - \alpha$. However, exact distribution-free conditional coverage $\mathrm{P}(Y_{n+1} \in \hat{C}_n \mid X_{n+1} = x) = 1 - \alpha$ almost surely for non-atomic continuous distributions is impossible in finite samples without producing infinite interval length (Vovk, 2012; Barber et al., 2021). CQR provides an empirical framework that adapts interval width according to estimated conditional quantiles. When base quantile regression estimators are consistent for the true conditional quantiles (which requires structural smoothness conditions on the base learner and data distribution; Meinshausen, 2006), the calibration adjustment $Q_{1-\alpha}$ converges in probability to zero, asymptotically recovering conditional validity. In the absence of base-model consistency, CQR provides guaranteed marginal coverage, with interval width responding empirically to estimated local conditional dispersion.

C. Weighted Conformal Prediction under Covariate Shift

When exchangeability fails due to covariate shift—where $P_{\text{train}}(X) \neq P_{\text{test}}(X)$ while the conditional response distribution $P(Y \mid X)$ remains invariant—Tibshirani et al. (2019) demonstrated that Weighted Conformal Prediction restores exact finite-sample coverage, provided that the exact covariate likelihood ratio $w(x) = dP_{\text{test}}(x) / dP_{\text{train}}(x)$ is known:

$$p_i(X_{n+1}) = \frac{w(X_i)}{\sum_{j=1}^{n_2} w(X_j) + w(X_{n+1})}.$$

If the density ratio $w(x)$ is unknown and must be estimated from finite samples (e.g., via logistic regression or kernel density estimation), the exact finite-sample coverage guarantee

is relaxed to an asymptotic guarantee or bounded by the estimation error $\|\hat{w} - w\|$.

D. Empirical Limitations & Dataset Considerations

The findings of this study should be interpreted alongside several methodological considerations. First, while $M = 100$ Monte Carlo replications provide sufficient statistical power to detect over-coverage under parametric misspecification ($p < 10^{-15}$), larger replication counts ($M \geq 1{,}000$) would further tighten standard errors around conformal coverage. Second, split conformal methods require withholding a portion of the data for calibration, reducing the sample size available for training the predictive model. Third, repeated random splits on the `MASS::Boston` dataset share data across partitions, meaning evaluations across splits are statistically dependent (Dietterich, 1998; Nadeau & Bengio, 2003); results on this historical benchmark serve as descriptive illustrations of stability rather than formal independent-sample tests. Evaluating larger modern high-dimensional tabular datasets represents a valuable direction for future research.

VI. Conclusion

This study compared classical parametric, bootstrap, Bayesian, and distribution-free conformal methods for uncertainty quantification in machine learning. Through Monte Carlo simulations and empirical evaluations in R, we demonstrated that while parametric OLS, bootstrap, and Bayesian Normal-Inverse-Gamma models achieve optimal efficiency under Gaussian homoscedasticity, they exhibit systematic positive coverage bias under likelihood misspecification due to outlier inflation of sample variance. In contrast, Split Conformal Prediction and CQR provide finite-sample marginal coverage guarantees under exchangeability, maintaining empirical coverage consistent with nominal 90% across all noise regimes. Conformalized Quantile Regression further adapts interval width to local heteroscedasticity, reducing proper Winkler scores relative to fixed-width conformal bands. Overall, conformal prediction establishes a reliable, distribution-free safety net for machine learning models, trading a modest calibration budget for guaranteed marginal predictive coverage.

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